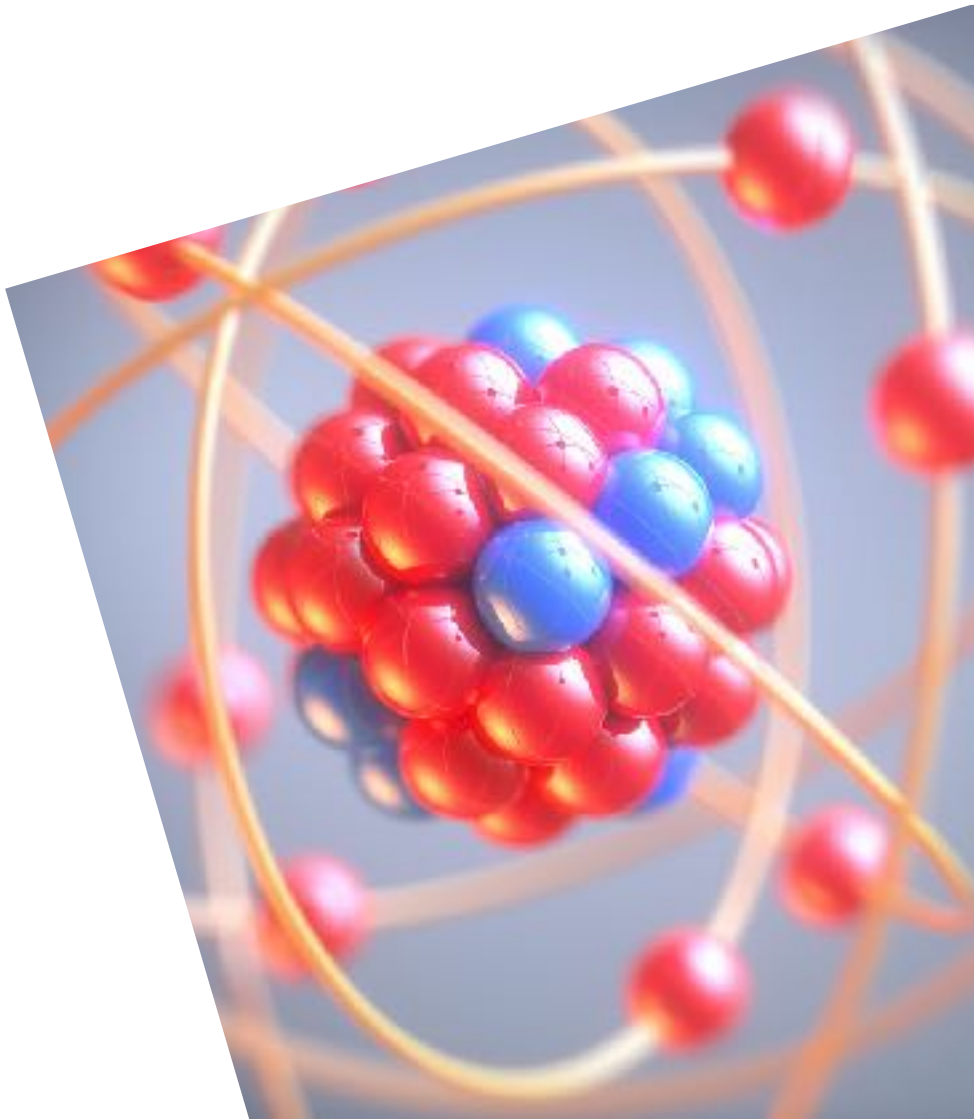
PHYC10009 Physics Foundations

Modern physics: atomic physics examples, the nucleus
Week 10 Lecture 3

Professor Phillip Urquijo
School of Physics
University of Melbourne

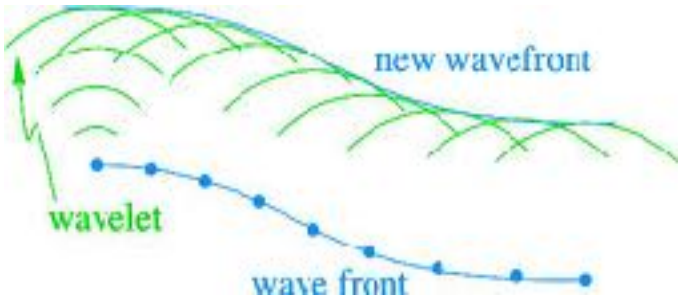


Recap

DeBroglie Wavelength

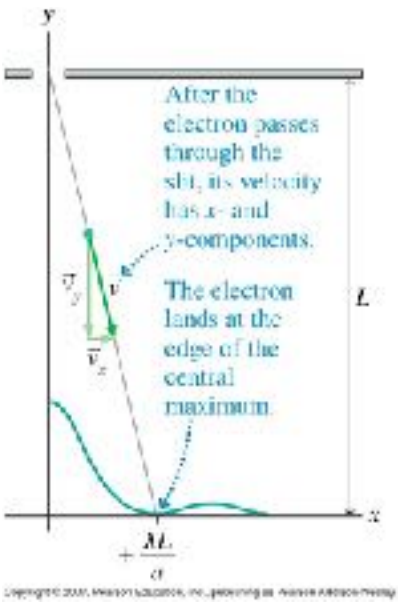
$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Huygens' Principle



the new position of the wavefront is a surface tangent to the the wavelets

Diffraction

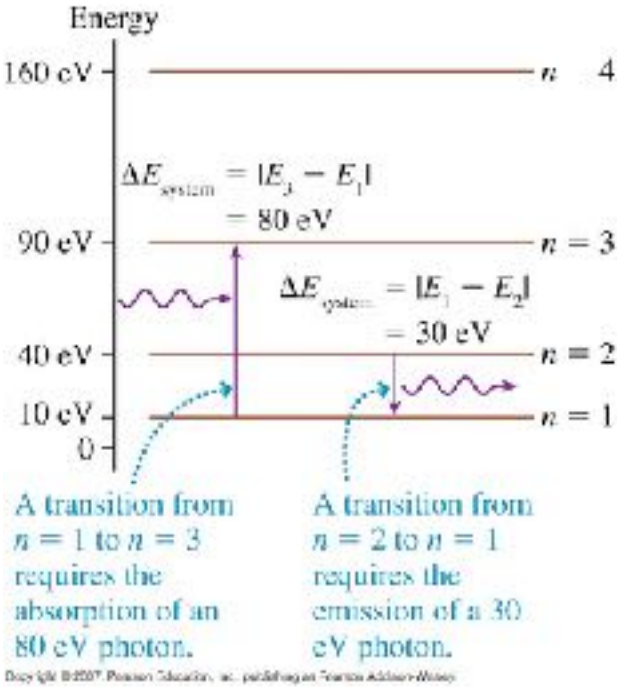


Uncertainty Principle

$$\Delta x \Delta p_x \geq \frac{h}{2\pi}$$

$$\Delta t \Delta E \geq \frac{h}{2\pi}$$

Energy is Quantised



Example question

A scientist measures the emission spectrum of hydrogen. Her spectrometer uses a silicon-based detector that only detects light from 400 – 1100nm. She finds emission at wavelengths of approximately 435, 488, 658, 1008, 1097nm.

Identify the principle quantum numbers n for the hydrogen energy levels for each of these wavelengths.

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Identify the principle quantum numbers n for the hydrogen energy levels for each of these wavelengths.

$$E = hf = \frac{hc}{\lambda}$$

$$h = 4.14 \times 10^{-15}\ eVs^{-1}$$

$$c = 3 \times 10^{8}\ ms^{-1}$$

So:

E(435 nm) = 2.9 eV $(5) \rightarrow (2)$

E(488nm) = 2.6 eV $(4) \rightarrow (2)$

E(658nm) = 1.89 eV $(3) \rightarrow (2)$

E(1008nm) = 1.24 eV $(7) \rightarrow (3)$

E(1097nm) = 1.13 eV $(6) \rightarrow (3)$

Hydrogen Energy Levels: $E_n = -\frac{13.60}{n^2}\ eV$

E(1) = -13.60 eV		
E(2) = -3.4 eV	To (2)	
E(3) = -1.511 eV	1.89 eV	To (3)
E(4) = -0.85 eV	2.6 eV	0.67 eV
E(5) = -0.544 eV	2.9 eV	0.97 eV
E(6) = -0.378 eV	3.02 eV	1.13 eV
E(7) = - 0.278 eV	3.12 eV	1.24 eV

Example question

The de Broglie wavelength for a particle is $\lambda = h/p$. The smallest particle we can see with an optical microscope is about 1 μm diameter. Assuming the particle density is that of water (1000 kg/m^3), calculate the mass and determine the speed of the particle such that its de Broglie wavelength is the same as its diameter.

Example question

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We have $\lambda = \dfrac{h}{p}$ and $p = mv$ so $\lambda = \dfrac{h}{mv} \Rightarrow v = \dfrac{h}{m\lambda}$

We know $\lambda = 1 \times 10^{-6} m$, $h = 6.63 \times 10^{-34}\ \boldsymbol{J}.s$ so we need to calculate m, the mass.

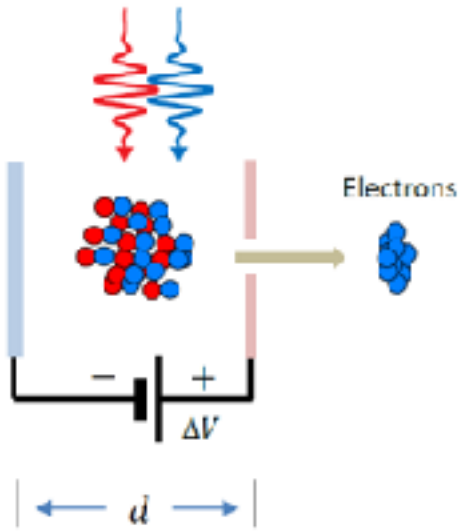
$m = \rho V$, where ρ is the density= $1 \times 10^{3} kg\ m^{-3}$and V is the Volume
$V = \dfrac{4}{3}\pi r^3$ where r is the radius of a sphere and $r = 0.5 \times 10^{-6} m$.

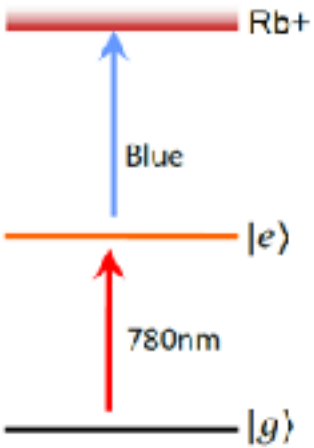
Plugging in the numbers we have: $V = 5.2 \times 10^{-19} m^3$ and so $m = 5.2 \times 10^{-16} kg$

So the velocity $v = \dfrac{6.63 \times 10^{-34}}{1.0 \times 10^{-6} \times 5.2 \times 10^{-16}} = 1.3 \times 10^{-16} ms^{-1}$

2013 exam

In the Scholten Lab, rubidium atoms are laser cooled to a temperature of 70μK. Laser light is used to excite and photoionise the cold atoms, releasing electrons. The process is very much like the photoelectric effect.





(a) The energy need to photoionise a Rb atom from the ground state $|g\rangle$ is 4.18 eV. What is the wavelength λ_{UV} of a single photon that will photoionise a ground state Rb atom?

(b) Lasers with this wavelength are not readily available, and instead Scholten uses two lasers, one at $\lambda_{780} = 780$ nm to excite the ground state atoms to state $|e\rangle$, and then a blue laser to ionise the excited atoms (see figure). What is the largest wavelength of blue light, λ_{blue}, that will ionise the excited atoms?

(c) If the wavelength of the blue laser is shorter, electrons will still be generated. Write an expression for the excess energy in terms of the three wavelengths above, and explain what happens to that excess energy.

(d) The electrons are accelerated by the electric field between two electrodes separated by distance $d = 4.0$ cm with potential difference $\Delta V = 4$ kV. What is the deBroglie wavelength of the electrons when they exit through an aperture in the positive electrode? Be careful: the atoms are equidistant from the two electrodes.

2013 exam solutions

$$h = 4.14 \times 10 - 15\ eV.s$$
$$h = 6.63 \times 10^{-34} J.s$$

(a) Photon Energy $E = hf$, and $\lambda f = c \Rightarrow\ E = \dfrac{hc}{\lambda}$

$$\Rightarrow \lambda = \frac{hc}{E} = \frac{4.14 \times 10^{-15} \times 3 \times 10^{-8}}{4.18} = 297\ \text{nm}$$

(b) Energy of state "e" $E_e = \dfrac{hc}{\lambda} = \dfrac{4.14 \times 10^{-15} \times 3 \times 10^{8}}{7.8 \times 10^{-7}} = 1.59eV$. So need 4.18-1.59=2.59eV.

$$\lambda = \frac{hc}{E} = \frac{4.14 \times 10^{-15} \times 3 \times 10^{8}}{2.59 \times 10^{-7}} = 479\ nm$$

(c) Let Energy for ionisation = E_{ionize}, Energy of "blue" light = E_{blue}

Energy of liberated electron = E_{lib}

$$E_{lib} = E_{blue} + E_e - E_{ionize} = \frac{hc}{\lambda_{blue}} + \frac{hc}{\lambda_{780}} - \frac{hc}{\lambda_{297}}$$

The excess energy is the kinetic energy of the liberated electron

(d) Equidistance from plates => electrons accelerated over half the distance so across half the potential difference.

$E_e = QV = 2.0 \times 10^3 eV$

$$\lambda = \frac{h}{p},\ p = \sqrt{2mE_e} = \sqrt{2 \times 9.1 \times 10^{-31} \times 2 \times 10^3 \times 1.6 \times 10^{-19}} = 2.41 \times 10^{-23}\ kg\ ms^{-1}$$

So $\lambda = \dfrac{6.63 \times 10^{-34}}{2.41 \times 10^{-23}} = 2.75 \times 10^{-11} m$

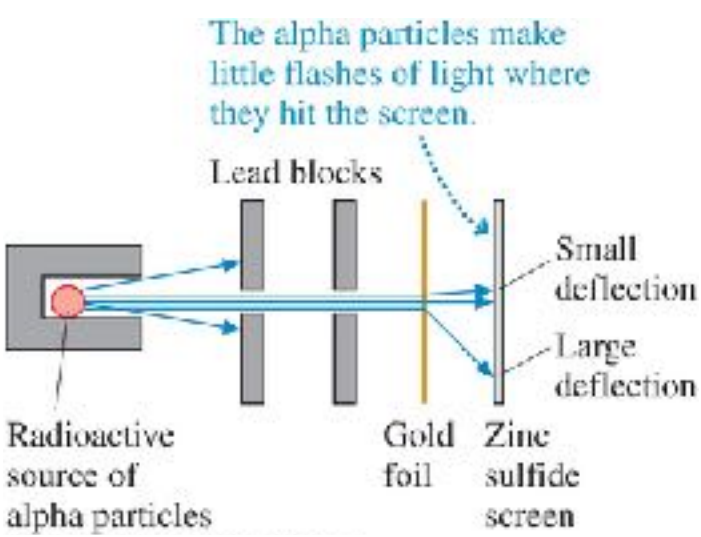
Today's Lecture

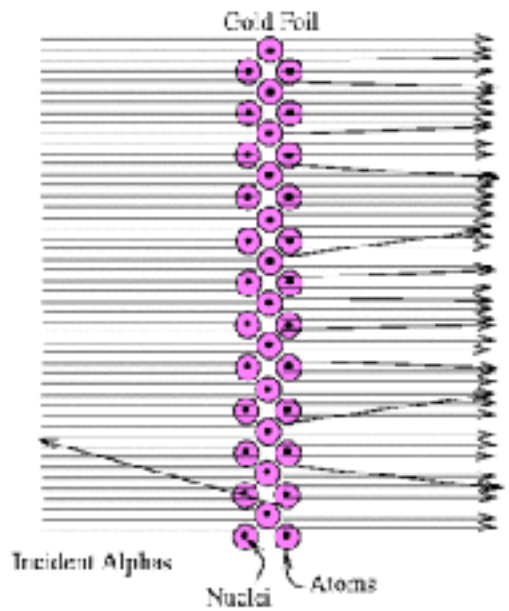
30.1 Nuclear structure

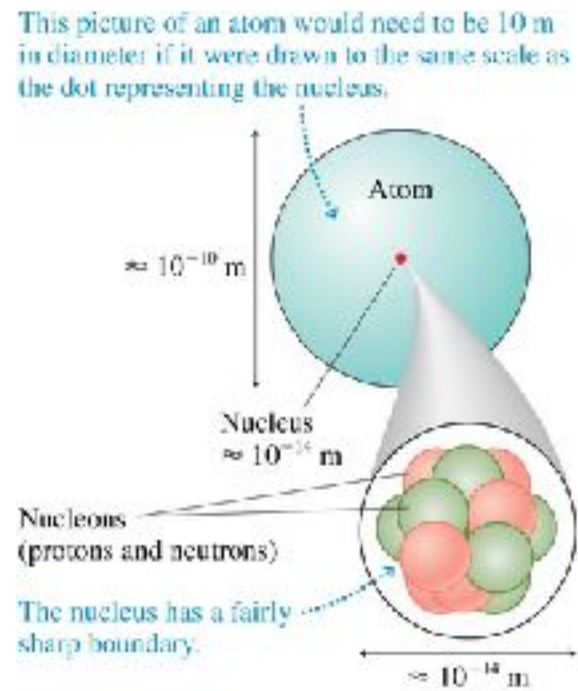
30.2 Binding energy and nuclear forces

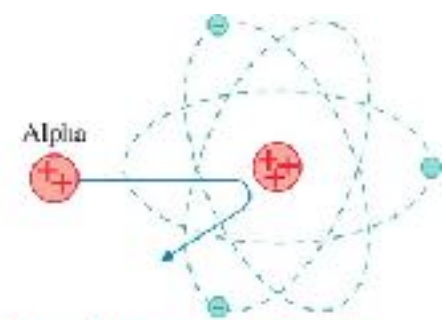
Atomic nucleus

- At the centre of an atom is a tiny nucleus
  - the size of an atom is the size of the electron cloud
  - nucleus is like pinhead at centre of cathedral
  - mostly empty space; 10^{-14}m *vs* 10^{-10}m
- Yet 99.95% of the ***mass*** of the atom is in the nucleus
- Geiger and Marsden (a kiwi), 1909
  - Scattering alpha particles





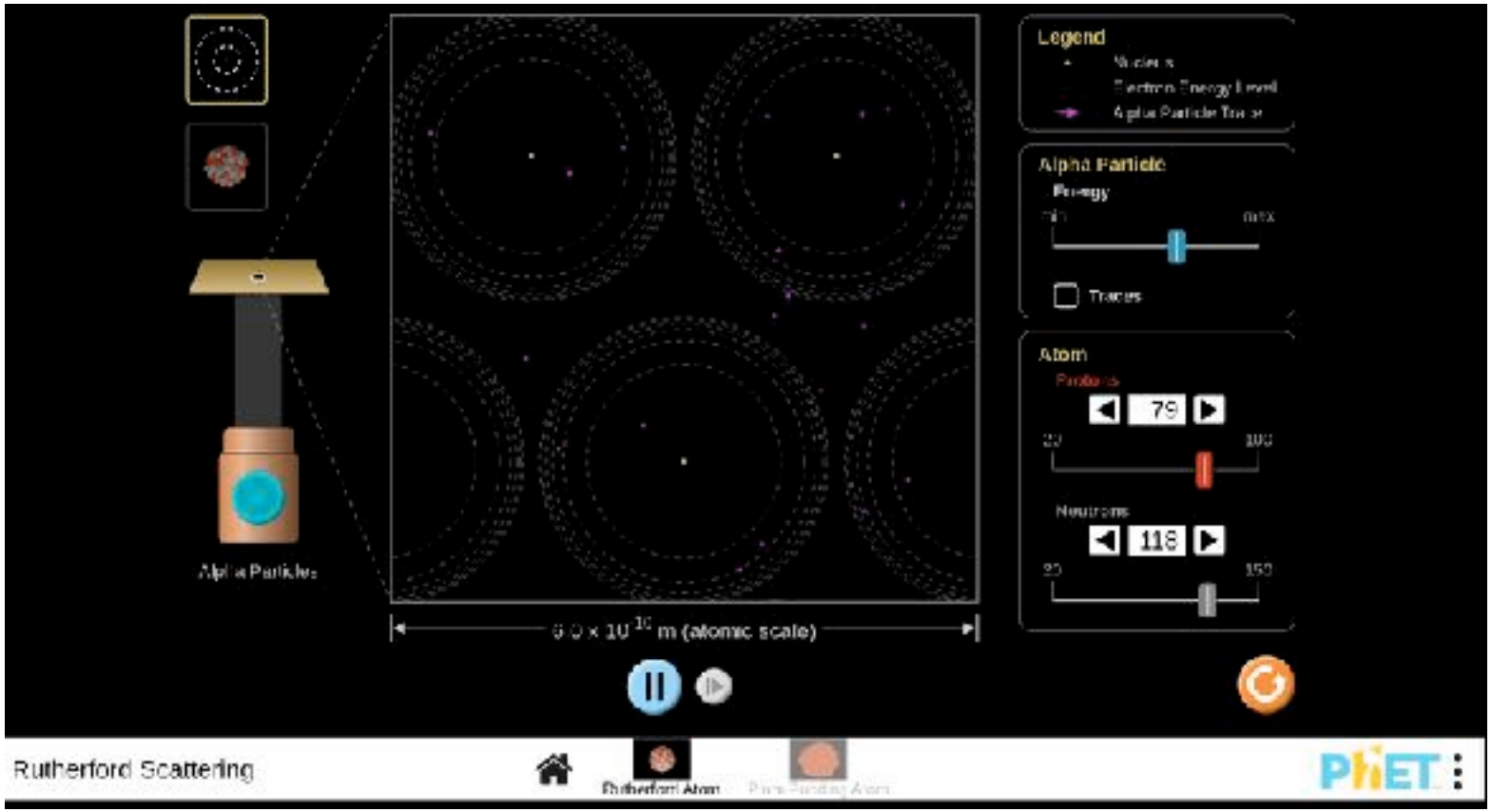




If the atom has a concentrated positive nucleus, some alpha particles will be able to come very close to the nucleus and thus feel a very strong repulsive force.

D Na-4 Rutherford Scattering

Simulation of Rutherford Scattering



https://phet.colorado.edu/sims/html/rutherford-scattering/latest/rutherford-scattering_en.html

Atomic Nucleus

Nucleus = particles called ***nucleons***

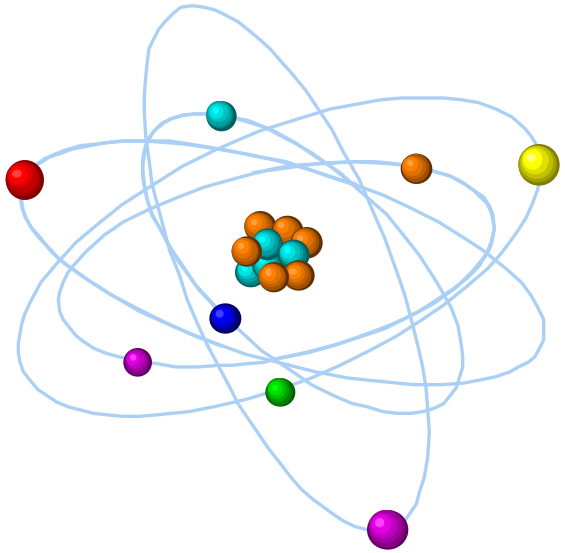
Protons: positive charge, number Z

Neutrons: neutral, number N

Total mass number $A = Z + N$

	Proton	Neutron
Number	Z	N
Charge	+e	0
Spin	1/2	1/2
Mass, in u	1.00728	1.00866

$$1\ u \equiv 1\ \text{amu} = \frac{1}{12}^{th} \text{ mass of } {}^{12}\text{C}$$



Nuclear isotopes

- Z = atomic number = proton number; defines the chemical element
  - E.g. hydrogen, $Z = 1$; helium $Z = 2$; …uranium $Z = 92$
  - Chemical behaviour depends on number of *electrons*
- N = neutron number
- Different N for same Z? Different number of neutrons: ***nuclear isotopes***
- Symbol: $^{A}_{Z}\mathrm{X}$ but note Z is redundant so often see $^{A}\mathrm{X}$

Examples

$^{16}_{8}\mathrm{O}$ and $^{17}_{8}\mathrm{O}$ Z same (both oxygen); different N (8 and 9 neutrons)

$^{85}\mathrm{Rb}$ and $^{87}\mathrm{Rb}$ Naturally found 72% and 28%

Neither is radioactive

means 87 grams per mole

Nuclear isotopes

- You are breathing nuclear isotopes!
- Many elements have several stable isotopes
  - Oxygen 99.76% ^{16}O 0.038% ^{17}O and 0.2% ^{18}O
  - Fractions are called the natural abundance of each isotope
- Chemically identical, hence always found in natural abundance

- *Not quite!*
  - Nuclear and other processes can affect abundance
  - Cosmic rays affect carbon-12 and carbon-14 abundance; basis of carbon dating
  - ^{16}O and ^{18}O ratios related to prehistoric Earth temperatures

Radioactive decay

Many isotopes are not stable: they are *radioactive*

Examples: ^{14}C, ^{40}K both common in biosphere

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Periodic Table

Elements of the periodic table are distinguished by

(a) The number of protons in the nucleus

(b) The number of neutrons in the nucleus

(c) The number of electrons in the atom

(d) Both (a) and (b)

(e) (a), (b) and (c)

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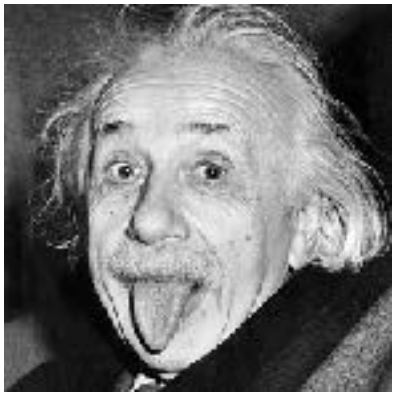
Rest mass



- A familiar equation: $E = mc^2$

- Matter can be converted to energy and v.v.
- A particle has energy even when at rest

	Rest mass energy
Electron	511 keV
Proton	938.28 MeV
Neutron	939.57 MeV

- One atomic mass unit, $1\ u = 1.6605 \times 10^{-27}\text{kg}$

- Energy/mass units?

$$E = mc^2 = 1.6605 \times 10^{-27}\left(2.9979 \times 10^8\right)^2 = 1.49 \times 10^{-10}\text{J}$$

$$\frac{1.49 \times 10^{-10}}{1.6 \times 10^{-19}} = 931.49\ \text{MeV}$$

30.2 Atomic rest mass

- Hey, wait a minute… Something doesn't add up!
- $1u$ = 1/12th mass of ^{12}C = 6 x proton + 6 x neutron
  - Proton mass 938.28 MeV/c^2
  - Neutron mass 939.57 MeV/c^2
  - So expect $1u$ =(^{12}C/12) = (6 x 938.28 + 6 x 939.57)/12 = 938.93 MeV/c^2

- But we measure ^{12}C/12 and find $1u$ = 931.49 MeV/c^2

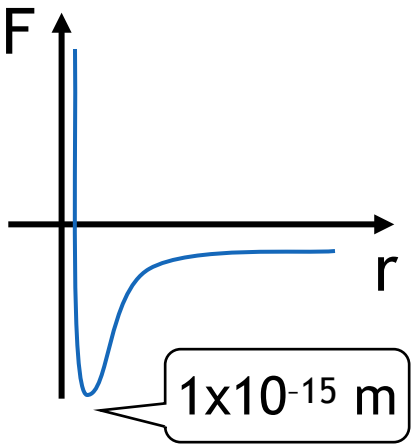
- Protons and neutrons are "lighter" within a nucleus than when free!

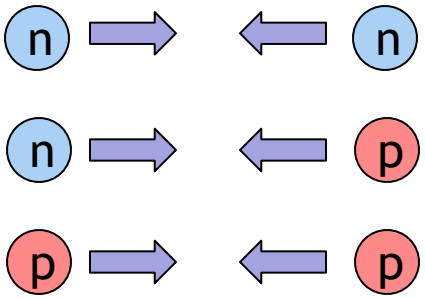
- Binding energy: the potential energy of the bond between nucleons

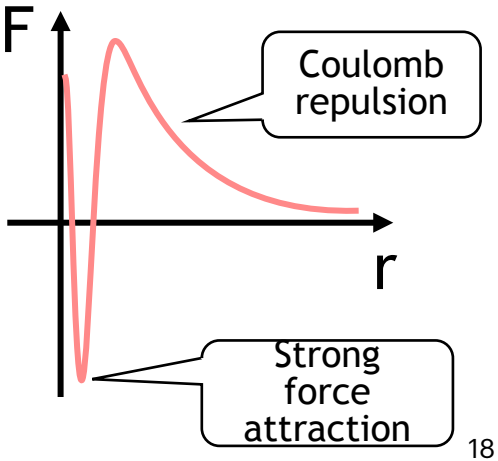
If we put some nucleons together, the mass of the combined nucleus is less than the separate nucleons. The difference is released as some other energy (e.g. gamma rays or kinetic energy)

Forces and energy in the nucleus

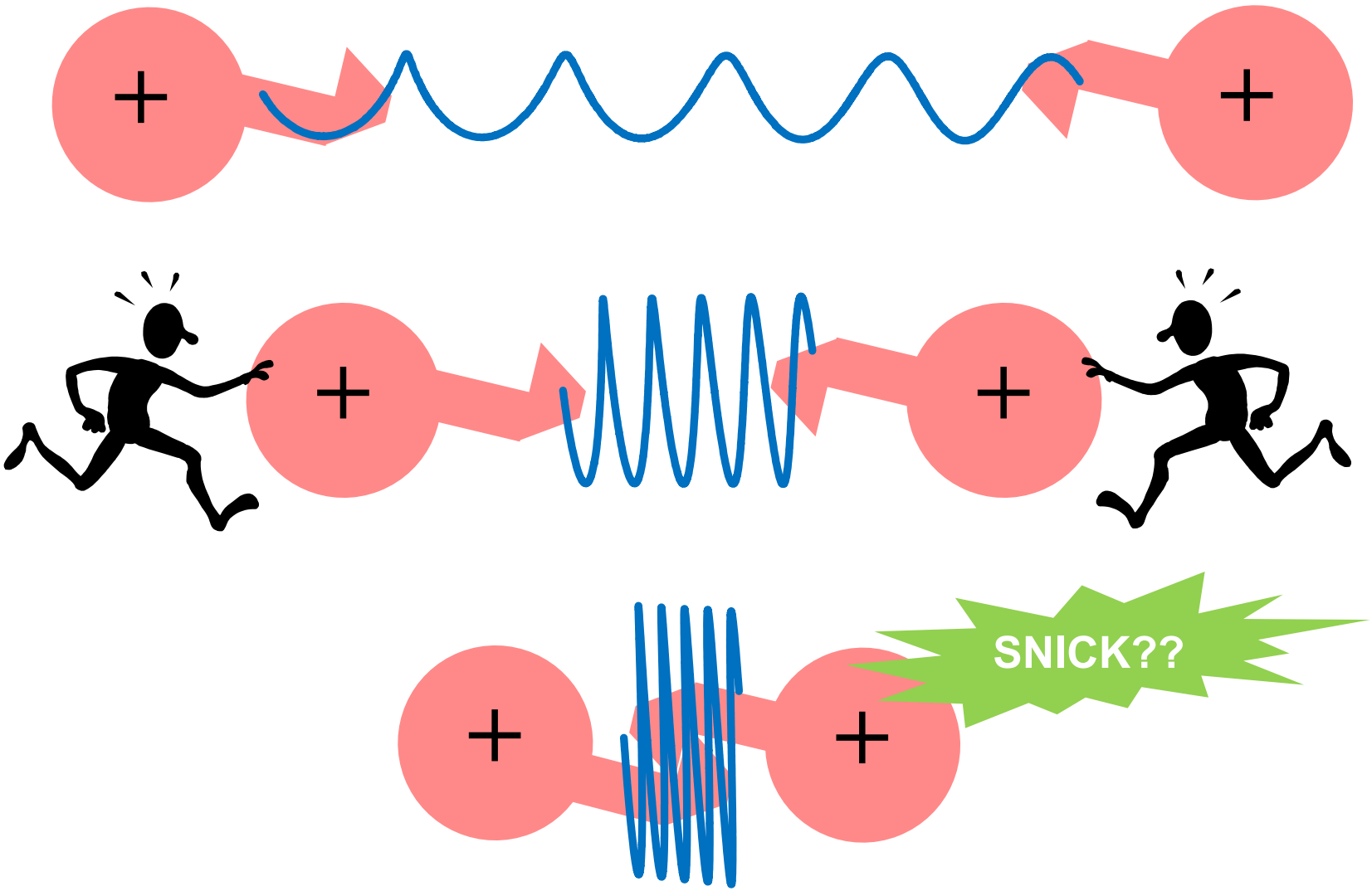
- Nucleons are bound
- Protons repel (Coulomb) – very strongly
- Something is holding them together:
  - Does not act on electrons
  - Very short range
  - Stronger than Coulomb at short range
  - Attractive for p-p, p-n and n-n
- Coulomb is $\propto \frac{1}{r^2}$ thus very strong at short range
- Nuclear force even stronger!



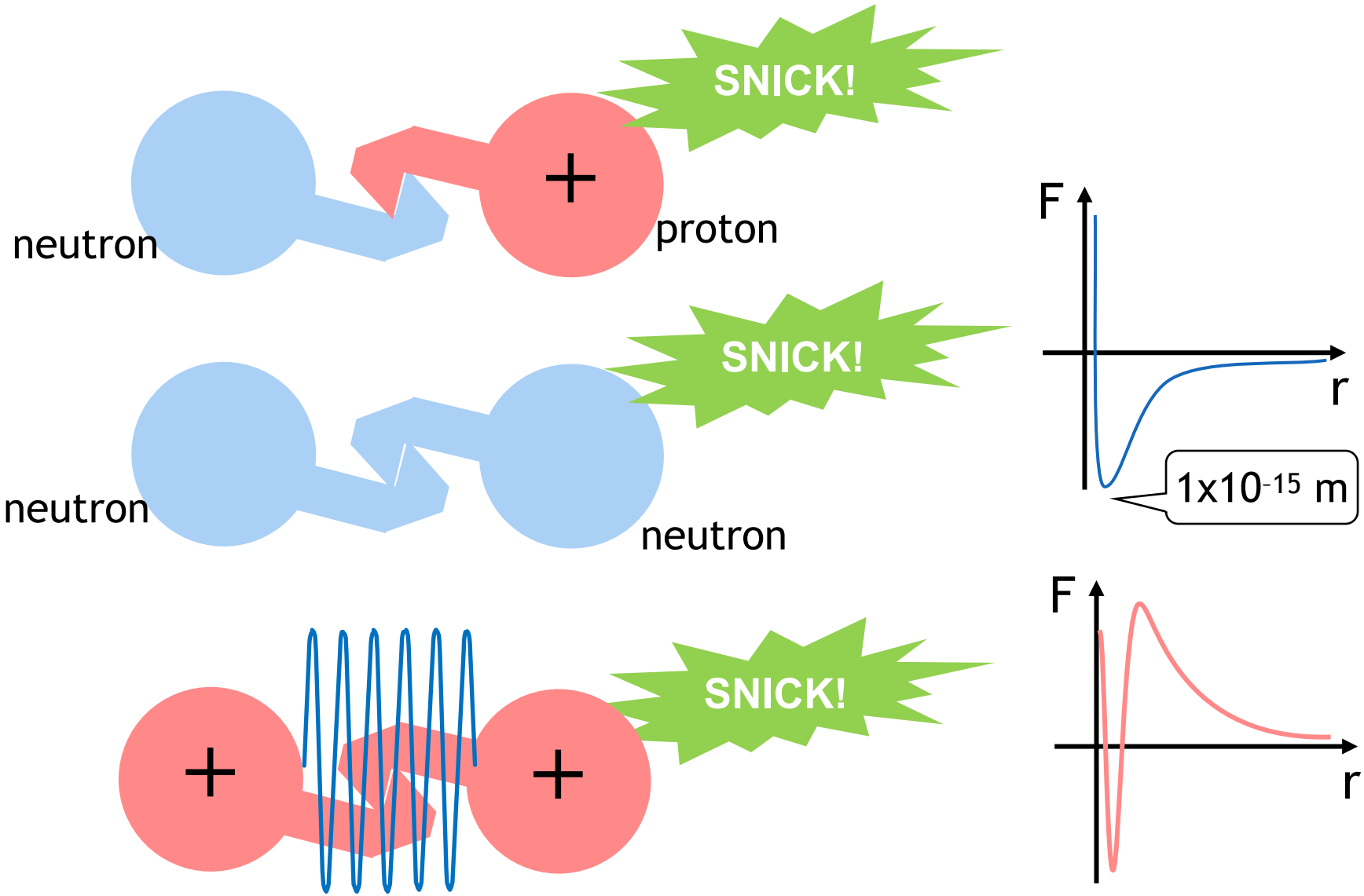




STRONG NUCLEAR force is short range



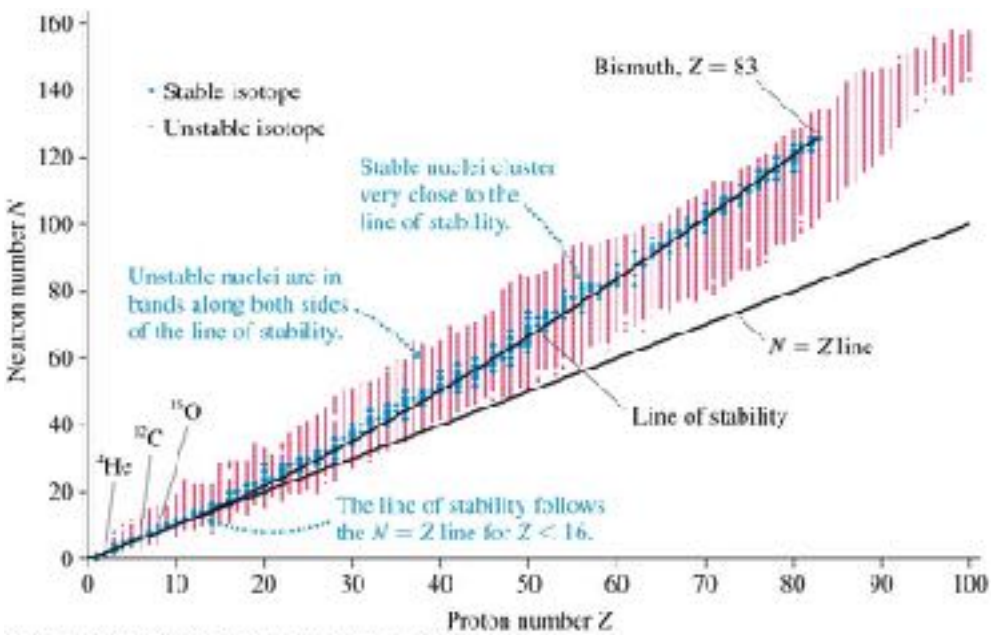
STRONG force also works for neutrons and protons



Nuclear stability

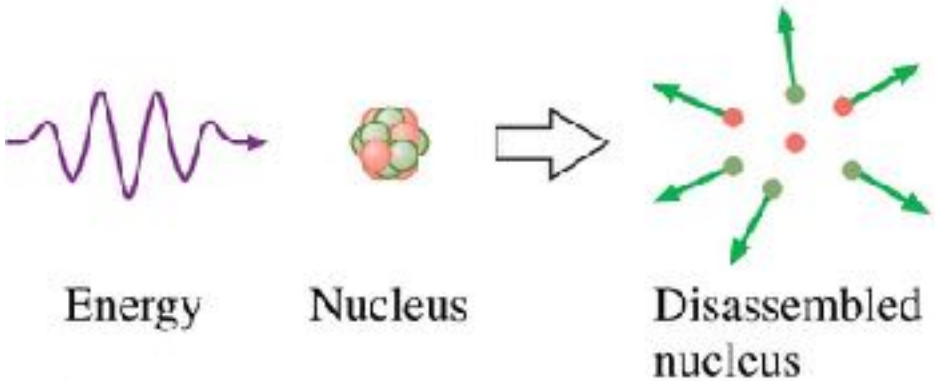
- Protons in nuclei <u>strongly</u> repel
- Nuclear strong force binds, but not strongly enough
- Neutrons: strong force without Coulomb repulsion
- Lightest elements $Z < 16$ stable if $N = Z$
- Larger nuclei need extra neutrons to increase strong-force binding

- No stable nuclei with $Z > 83$ (bismuth)
- Naturally occurring nuclei up to $Z = 92$ (uranium) but unstable



Binding energy

- Binding energy = energy required to liberate bound neutrons and protons
- Difference between separate, free nucleons and combined nucleus
- Binding energy is not something a nucleus *has* – it is energy it "lacks" relative to the total mass of its separate constituents.



- Binding energy

$$B = \left(Z\, m_H + N\, m_n - m_{\text{nucleus}}\right) \times 931.49 \text{ MeV/u}$$

$m_p =$ Hydrogen mass in u (1.0078 u)
$m_n =$ neutron mass (1.0087 u)
$Z =$ number of protons
$N =$ number of neutrons
$m_{\text{nucleus}} =$ mass of nucleus (as a neutral atom)

Question

How much energy to separate all the protons and neutrons in ^{12}C?

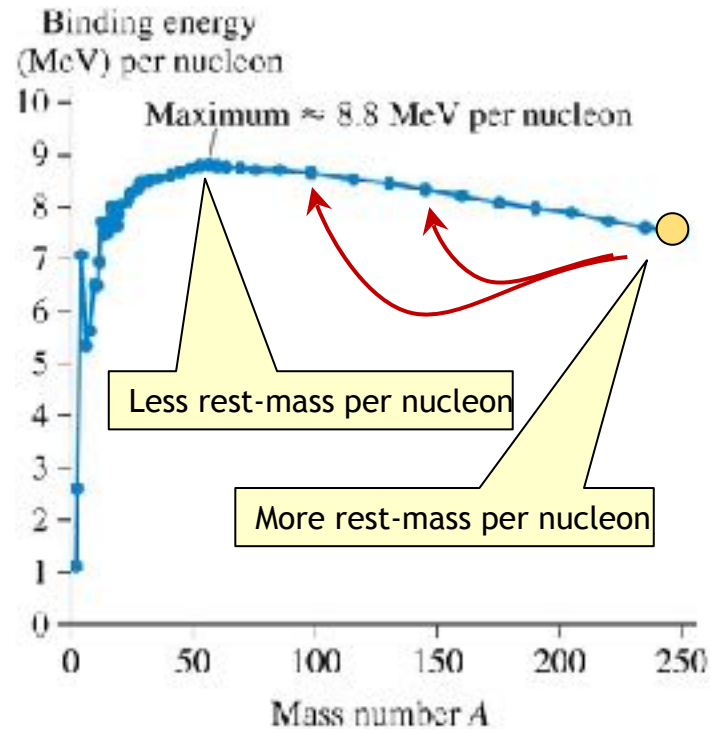
$$B = \left(6 \times 1.0078u + 6 \times 1.0087u - 12\right) = 0.099$$

$$B = 0.099 \times 931.49 = 92.23 \text{ MeV}$$

That is a *lot* of energy in a single ^{12}C nucleus…

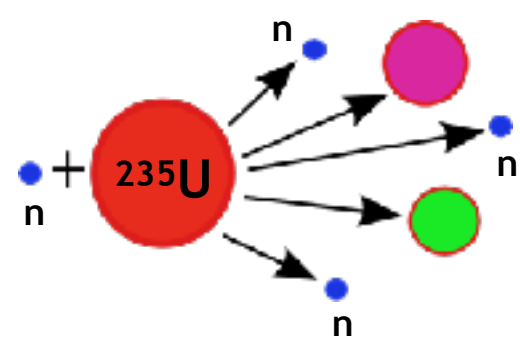
Binding energy

- How much energy is this 92.23 MeV?
- Metabolic energy in eukaryotic cells is ATP
  - 0.625 eV per ATP molecule
  - $C_{10}H_{16}N_5O_{13}P_3$
- 16 carbon nuclei → *2 billion* × chemical energy
  - N, O and P all have even *more* binding energy
- Nuclear binding energy *per nucleon* increases until $A = 56$ then decreases



Fission

- Split ^{235}U nucleus into two fragments ^{140}Xe + ^{94}Sr + 2n
- Fragments more tightly bound, thus a small fraction of total binding energy *released*
- Energy released from one gram of uranium same as 5 tanks of petrol or 3 tonnes of coal
- 3MWh per gram and no CO2



200MeV

Binding energy

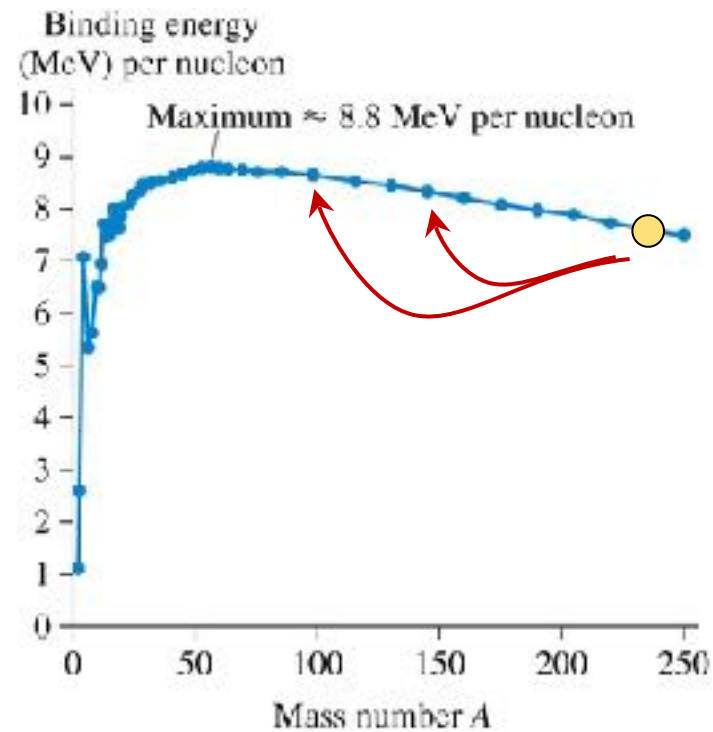
Fission

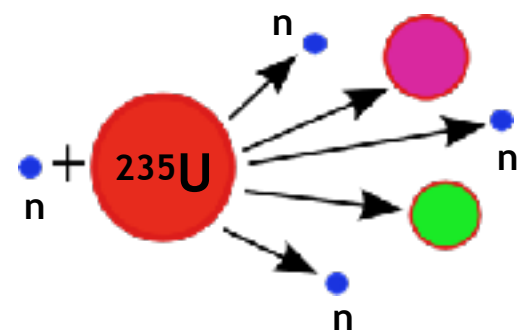
- Consider $n + {}^{235}\mathrm{U} \rightarrow {}^{140}\mathrm{Xe} + {}^{94}\mathrm{Sr} + 2n$
- Binding energy per nucleon, total:
  - 7.6 MeV per nucleon → 1794 MeV total
  - ^{140}Xe 8.4 MeV → 1176 MeV
  - ^{94}Sr 8.6 MeV → 808 MeV

  } 1984 MeV
- Fragments more tightly bound
- Binding energy is *negative*
- Thus energy released = (final) – (initial)

  1984 – 1794 = 190 MeV
- Lower energy at end; difference 190 MeV
- Energy released (kinetic energy of fragments)

- Think of hydrogen atom: electron falling from high state (small negative energy) to low state (energy more negative)

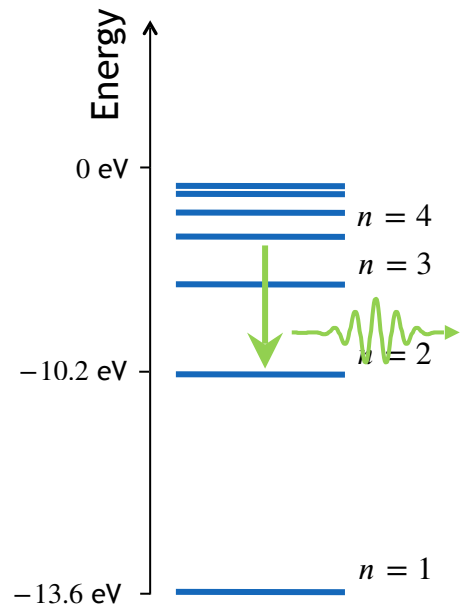




200MeV

Nuclear binding energy

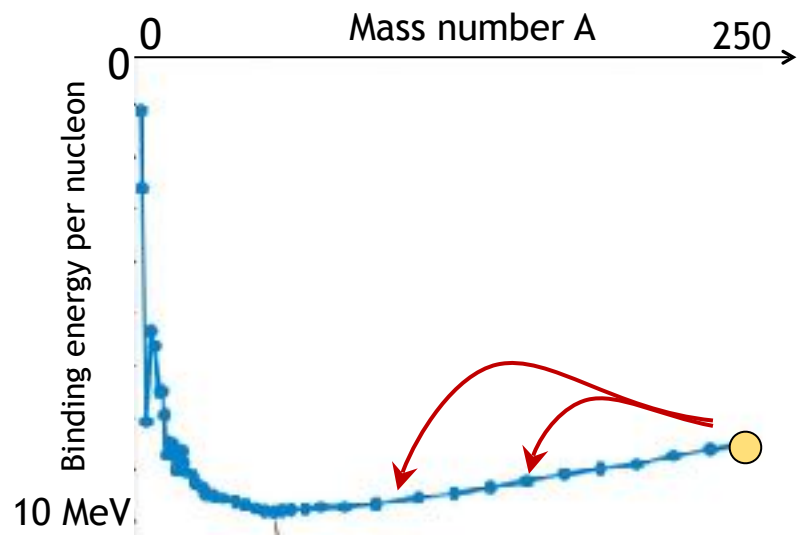
- Binding energy = energy required to liberate bound neutrons and protons
- Electron binding in hydrogen
  - Energy levels = energy to separate electron



- Nuclear
  - Fission: from high A to low A
  - Fusion: from low A to high A
  - In both cases, BE/nucleon increases (more negative)
  - Energy released

- Like dropping a weight: potential energy converts to kinetic

- Like a negative electron attracted to a positive charge



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Proton and Neutron

Which has greater mass?

(a) A neutron and a proton that are far apart (unbound)

(b) A neutron and a proton that are bound together in a hydrogen (deuterium) nucleus

(c) Both the same

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Energy in Nucleus

A nucleus has

(a) More energy than its component neutrons and protons

(b) Less energy than its component neutrons and protons

(c) The same energy as its component neutrons and protons

(d) More energy than its component neutrons and protons have when the nucleus is at rest but less energy than when it is moving.

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Summary

- Nuclei = nucleons = protons + neutrons $A = Z + N, \ {}^{A}_{Z}\mathrm{X}$

	Proton	Neutron
Number	Z	N
Charge	+e	0
Spin	1/2	1/2
Mass, in	1.00728	1.00866

	Rest mass energy
Electron	511 keV
Proton	938.28 MeV
Neutron	939.57 MeV

- Binding energy and $E = mc^2$

$$B = \left(Z\, m_H + N\, m_n - m_{\text{nucleus}}\right) \times 931.49 \text{ MeV/u}$$